

corresponding relays via relay driver IC3. When key 8 is pressed, IC2 resets, de-energising all the relays. It means all the appliances can be switched on independently one by one and switched off simultaneously.

Suppose you want to switch on the bulb (load 1). Lift the handset from the cradle and press key 1. Relay RL1 energises to switch on the bulb. The bulb continues to glow even when you put the handset back on the cradle. To

switch off the bulb (de-energise relay RL1), you'll have to again lift the handset and press key 8.

If you want to switch on all the seven appliances, lift the handset and press keys 1 through 7 in any sequence. Relays RL1 through RL7 will energise one by one in the order of pressing the keys. To switch off all the appliances at one go, just press key 8.

The circuit works off regulated 12V using either a 12V adaptor or a

12V battery.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Mount switches S1 and S2 on the front side of the cabinet. Switch S1 is used to provide the power supply to the circuit and switch S2 is used to disconnect the circuit from the telephone. Use relays with proper contact ratings to handle the load current. Connect the circuit to the telephone line. ●

Active Reflector for Pedestrian Safety

T.K. HAREENDRAN

Pedestrians crossing roads in badly-lit areas, particularly at night or in the early morning, run a risk of accident. There are ready-made arm bands available that reflect the light of incoming traffic to catch the attention of the drivers.

It is also possible to construct an active reflector to alert the drivers of a pedestrian crossing the road. Presented here is the circuit of an active reflector that can warn the drivers

even when no light falls on it. Even without the ambient light or vehicle headlight falling on its sensor, the LED blinks.

The circuit (Fig. 1) consists of a light-dependent resistor (LDR1), timer IC NE555 (IC1), transistor BC547 (T1) that acts as a buffer and LED driver, and a bright red LED (LED1). NE555 timer is configured in oscillator mode. When ambient light is dim, there is a potential drop across the light-dependent resistor below the level set by the combination of VR1 and R1,

and reset terminal (pin 4) of NE555 goes high to enable the oscillator.

Transistor BC547 (T1) then switches LED1 on and off at the oscillation frequency. Resistor R3 (220-ohm) limits the current through the LED to a safe value. Sensitivity of light sensor LDR1 can be set using potentiometer VR1.

After construction and testing, house the circuit in an appropriate cabinet with

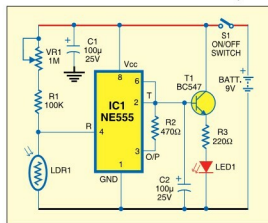


Fig. 1: Circuit of active reflector for pedestrian safety



Fig. 2: Pedestrian symbol

LDR1, VR1 and LED1 mounted on the front panel. Mount LDR1 such that LED light does not fall on it. Glue a piece of self-adhesive black sheet marked with a transparent pedestrian symbol to the front panel (refer Fig. 2). Next, attach a cloth peg at the rear side of the enclosure using a suitable screw and attach the reflector to the cloth with the cloth peg clip. ●

Dual-Colour Stroboscope

RAJU BADDI

Stroboscope is a device used to make a cyclically moving object appear slow-moving or station-

ary. This is realised by illuminating the object intermittently with short pulses of light. Stroboscope is used in the study of insect flight. It can also be used for experiments with simple

pendulum, studying details of rapidly moving objects and strobo-animation.